Understanding Safety Based on Urban Perception

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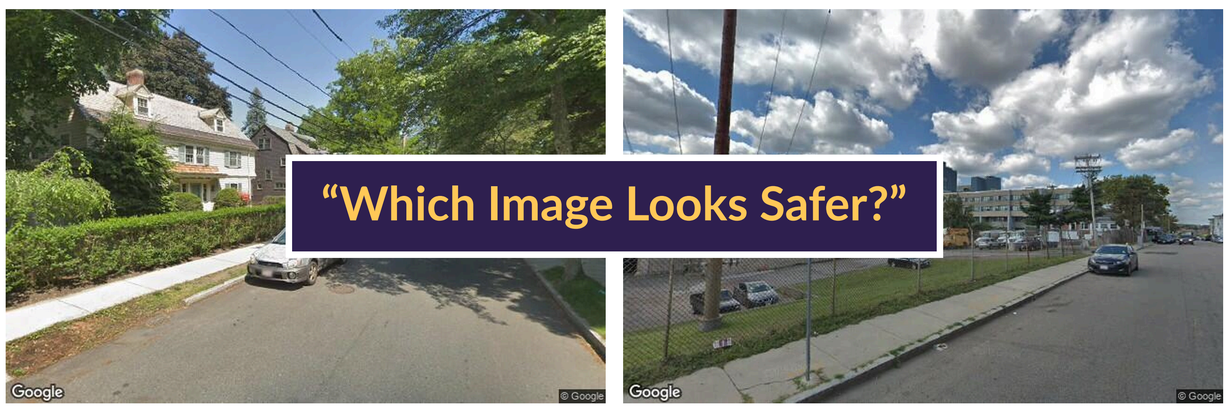
Motivation

- **Crime data:** many methods infer whether a street is dangerous from crime datasets or statistics, to predict criminal activity or map crime.
- **Human perception:** other works relate perceived safety to the visual appearance of streets, and train models that score safety from street photos.
- **The gap:** these models output a score, but not the reason behind it.

If an image is classified as safe, what visual features make the model predict this score?

Dataset

Google Street View images from **MIT Place Pulse 1.0**, with safety scores calculated from human answers comparing two images.



4,132	2	2
images grouped by city	cities: New York and Boston	classes: safe and not safe

Methodology

Street photo — Google Street View image with a human safety score ▸ **VGG16** — Fine-tuned network classifies it as safe or not safe ▸ **LIME** — Shows which regions drive the prediction

Image classification

We adapted **VGG16**, trained on ImageNet (millions of images, 1,000 classes), then fine-tuned it via back-propagation. One model per city.

Model explanation

LIME is a local, model-agnostic technique that explains a black-box model around one prediction:

- It simulates local candidates close to the original prediction.
- It uses L2 distance to the original as "local fidelity".
- It selects the features for the explanation with its Submodular Pick algorithm (SP-LIME).

Experiments

60% Training	20% Testing	20% Validation

64	100	0.0001	SGD
batch size	epochs	learning rate	optimizer

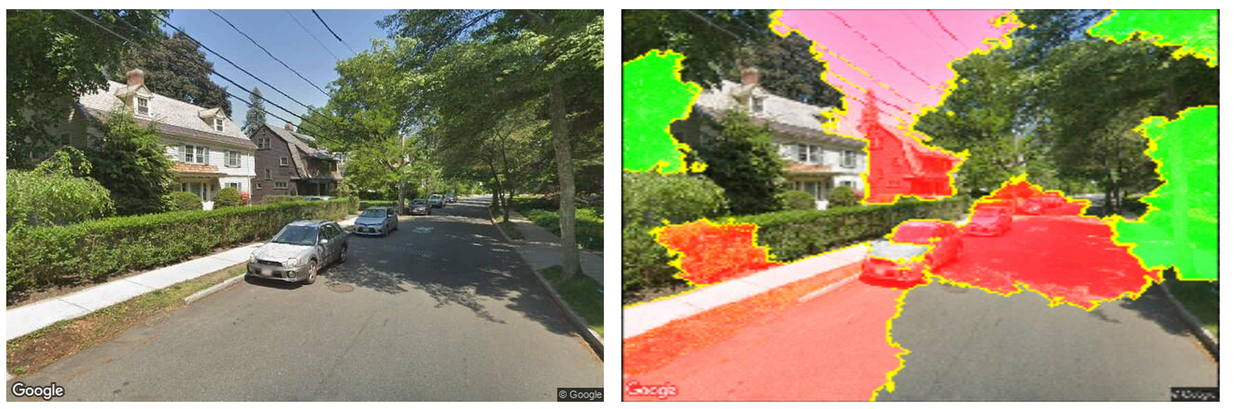
Testing accuracy

76%	69%
Boston	New York City

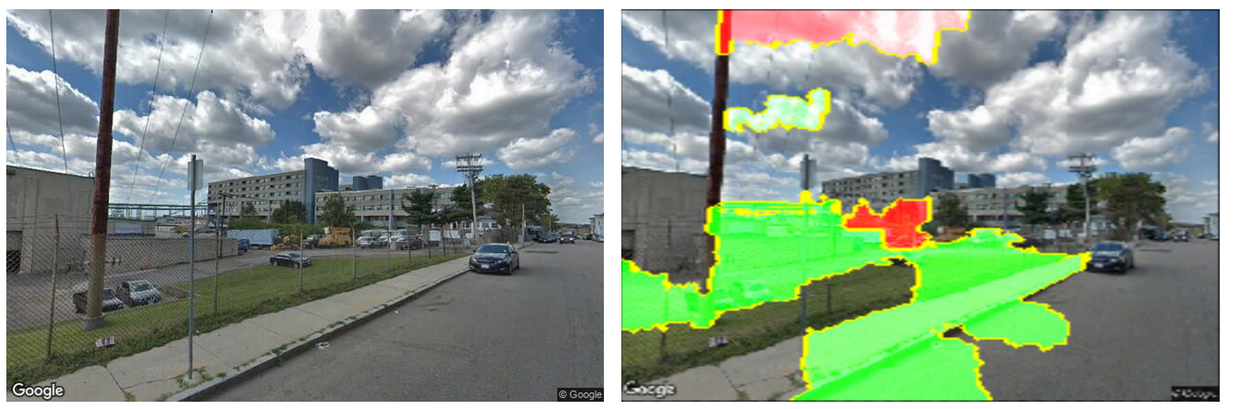
Explaining the Predictions

Pros: help predict the class — Cons: do not help the prediction

Boston, actual score 8.35, predicted: safe



Boston, actual score 1.06, predicted: not safe



- Both test images were **classified correctly**.
- In the safe image, **pros correspond to trees** and **cons to asphalt**.
- This lets us check whether the model associates scores with image features.

Next: run more experiments to verify whether green areas are more prevalent in "safe" images.

Acknowledgment & References

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